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Kelas : A

Creating and Observing a Simple Process

Bash dapat menciptakan proses baru dengan menjalankan perintah di background, selain itu setiap proses juga memiliki *PID* yang unik dan berbeda pada tiap prosesnya (kode yang sama dapat memiliki *PID* yang beda jika dijalankan ulang). Selain menciptakan proses baru, bash juga dapat menghentikan sebuah proses dengan perintah *kill*.

```
dewa-fitrah@dewa-fitrah-VMware-Virtual-Platform:~/hands-on-3$ ./task1_process_basic.sh
Running a sleep process in the background...
Background process PID: 5565
Displaying process status with ps:
  PID   PPID  STAT  CMD
  5565   5564  S+    sleep 30
Waiting for 3 seconds...
Checking process status again:
  PID   PPID  STAT  CMD
  5565   5564  S+    sleep 30
Stopping the process...
Status after kill:
  PID   PPID  STAT  CMD
```

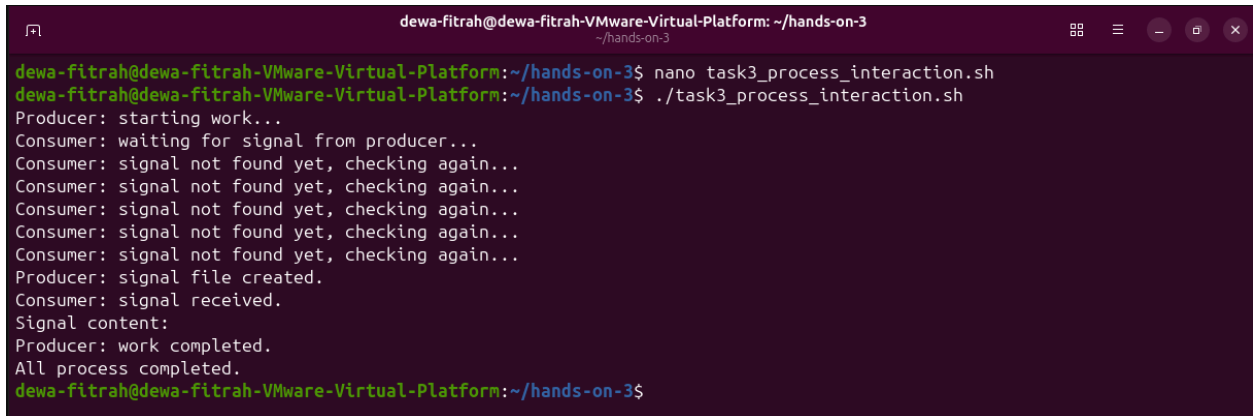
Monitoring the Status of Another Process

Perintah *kill -o* digunakan untuk memeriksa keberadaan proses. Looping digunakan untuk polling berkala.

```
dewa-fitrah@dewa-fitrah-VMware-Virtual-Platform:~/hands-on-3$ ./task2_check_process_status.sh
Starting worker process...
Worker PID: 5936
[13:45:35] Worker has finished.
Monitoring script finished.
dewa-fitrah@dewa-fitrah-VMware-Virtual-Platform:~/hands-on-3$ ./task2_check_process_status.sh
Starting worker process...
Worker PID: 5948
[13:46:19] Worker has finished.
Monitoring script finished.
dewa-fitrah@dewa-fitrah-VMware-Virtual-Platform:~/hands-on-3$
```

Interacting Between Process Using a Signal File

Fungsi *producer()* berjalan di background menuliskan *flag* selesai kepada *signal file*. Fungsi *customer()* juga berjalan di background dan melakukan pengecekan atau polling setiap 1 detik hingga file ditemukan. Setelah *signal file* terdeteksi, maka isi dari file tersebut akan ditampilkan pada terminal.

A terminal window titled 'dewa-fitrah@dewa-fitrah-VMware-Virtual-Platform: ~/hands-on-3' with standard window controls. The terminal shows a user running 'nano task3_process_interaction.sh' and then './task3_process_interaction.sh'. The script's output shows a producer starting work, a consumer waiting for a signal, the producer creating the signal file, the consumer receiving it, the producer completing work, and the consumer displaying the signal content before all processes complete.

```
dewa-fitrah@dewa-fitrah-VMware-Virtual-Platform: ~/hands-on-3$ nano task3_process_interaction.sh
dewa-fitrah@dewa-fitrah-VMware-Virtual-Platform: ~/hands-on-3$ ./task3_process_interaction.sh
Producer: starting work...
Consumer: waiting for signal from producer...
Consumer: signal not found yet, checking again...
Consumer: signal not found yet, checking again...
Consumer: signal not found yet, checking again...
Consumer: signal not found yet, checking again...
Consumer: signal not found yet, checking again...
Producer: signal file created.
Consumer: signal received.
Signal content:
Producer: work completed.
All process completed.
dewa-fitrah@dewa-fitrah-VMware-Virtual-Platform: ~/hands-on-3$
```

Observing the Process Lifecycle (start, running, stop, terminated)

Fungsi `cleanup()` berfungsi sebagai signal handler dengan perintah `trap`, sehingga saat sinyal diterima, proses tidak akan langsung mati, melainkan menjalankan `cleanup` terlebih dahulu. Setelah mencetak `PID`, script akan mencetak nomor iterasi (`count`) tiap 2 detik hingga terdapat perintah `kill` dari terminal kedua.

```
dewa-fitrah@dewa-fitrah-VMware-Virtual-Platform: ~/hands-on-3
Process started with PID 17506
Process is running... iteration 1
Process is running... iteration 2
Process is running... iteration 3
Process is running... iteration 4
Process is running... iteration 5
Process is running... iteration 6
Process is running... iteration 7
Process is running... iteration 8
Process is running... iteration 9
Process is running... iteration 10
Process is running... iteration 11
Process is running... iteration 12
Process is running... iteration 13
Process is running... iteration 14
Process is running... iteration 15
Process is running... iteration 16
Process is running... iteration 17
Process is running... iteration 18
Process is running... iteration 19
Process is running... iteration 20
Process is running... iteration 21
Process is running... iteration 22
Process is running... iteration 23
Process is running... iteration 24
Process is running... iteration 25
Process is running... iteration 26
Process is running... iteration 27
Process is running... iteration 28
Process is running... iteration 29
Process is running... iteration 30
Process received termination signal.
Performing cleanup before exiting...
```

```
dewa-fitrah@dewa-fitrah-VMware-Virtual-Platform:~$ ps -p 17506 -o pid,ppid,stat,cmd
  PID   PPID  STAT  CMD
  17506   6364  S+    /bin/bash ./task4_process_lifecycle.sh
dewa-fitrah@dewa-fitrah-VMware-Virtual-Platform:~$ kill -TERM 17506
dewa-fitrah@dewa-fitrah-VMware-Virtual-Platform:~$
```

Inter-Process Dependency in a Simple Workflow

Fungsi `download_data` dijalankan pada background, lalu main menunggu proses tersebut selesai sebelum lanjut ke tahap berikutnya. Setelah download selesai, `validate_data` memeriksa apakah file data ada (`-f`) dan tidak kosong (`-s`), lalu mengembalikan exit status 0 jika valid. Jika valid, `generate_report` dijalankan untuk menghitung jumlah baris data dan menyimpannya ke `/tmp/report.txt`. Urutan step 3 dan step 2 tercetak terbalik, merupakan interleaving output antar concurrent process di terminal.

```
dewa-fitrah@dewa-fitrah-VMware-Virtual-Platform: ~/hands-on-3
dewa-fitrah@dewa-fitrah-VMware-Virtual-Platform:~/hands-on-3$ nano task5_process_dependency.sh
dewa-fitrah@dewa-fitrah-VMware-Virtual-Platform:~/hands-on-3$ chmod +x task5_process_dependency.sh
dewa-fitrah@dewa-fitrah-VMware-Virtual-Platform:~/hands-on-3$ ./task5_process_dependency.sh
Waiting for download process to finish...
Step 1: Collecting data...
Step 1 completed: data saved in /tmp/sample_data.txt
Step 3: Generating report...
Step 2: Validating data...
Data is valid.
Report created in /tmp/report.txt
Workflow completed successfully.
Report content:
3 /tmp/sample_data.txt
dewa-fitrah@dewa-fitrah-VMware-Virtual-Platform:~/hands-on-3$
```